

JONNY LI

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Experience

SoundHound AI

Toronto, Canada

Staff Machine Learning Engineer

Aug 2026 – Present

- Introduced and led several major GPU/model efficiency initiatives across the company for ASR, LLM, TTS, training, and inference, leading to massively reduced latency/cost, improved product experience, and faster business execution.
- Designed and built production multi-modal LLM inference server with **2x faster first-token latency** and **1.2x faster per-token latency** compared to open source (vLLM, SGLang), using optimal scheduling, custom Triton/CUDA kernels, CUDA graphs, continuous batching, KV cache, and more.
- Reduced audio model pre-training time by **70%** on H100 GPU cluster using roofline analysis and networking/CPU/GPU optimization, saving \$100k USD per training run and massively increasing both R&D and business execution speed.

Senior Machine Learning Engineer

Aug 2025 – Jul 2026

- Built the fastest Qwen3-TTS realtime streaming inference server available today, with input streaming and sub **40ms** first audio latency, achieving **100x** cost savings and unlocking new roadmaps for modeling teams.
- Enabled text LLMs to become voice agents through designing and implementing a low-latency realtime streaming orchestration server for ASR, LLM, TTS models, optimizing all things from low-level model inference in PyTorch/CUDA to Python asyncio event loops, reducing end-to-end response latency from **2.5s to 220ms**.
- Authored LLM post-training library for H100 GPU cluster supporting mixture-of-experts models up to **120B** and long context training, using Huggingface, PyTorch, DeepSpeed, Docker, Kubernetes, and Ray.
- Conducted research experiments in LLM post-training modeling, evaluation methodology, and designing/implementing simulation and synthetic data generation pipelines.

Machine Learning Engineer II

Aug 2023 – Jul 2025

- Led model training/experimentation of LLM-based generative error correction for ASR, utilizing synthetic data generation, distillation, and loss function modifications to achieve **+90%** accuracy on prompted entity names.
- Built hyperparameter tuning pipeline for tuning dataset mixture weights, delivering **+30%** accuracy improvement in production ASR model.

Software Engineer

Oct 2021 – Jul 2023

- Engineered large-scale Spark ETL pipeline handling tens of TBs of data, debugged out-of-memory issues, and implemented salting/sharding algorithms to resolve data skew, achieving **2x** throughput of previous pipeline and significantly reducing experiment iteration time.
- Authored end-to-end MLOps pipelines, designed clear and extensible APIs, resolved complex Docker dependency issues, and setup CI/CD for end-to-end testing, reducing experiment iteration time from **2 days to 10 hours**.

Amazon

Vancouver, Canada

Software Engineer Intern

Jun 2020 – Aug 2020

- Built Python dependency graph analyzer to remove redundant configs, optimizing backend search engine efficiency.

Mitsucari

Tokyo, Japan

Software Engineer Intern

Sep 2018 – Aug 2019

- Full stack web, Rails, PostgreSQL, Heroku, jQuery, Bootstrap, SASS.

Projects (Github)

Open Source Contributor | *Hugging Face Transformers, TRL, PEFT*

2024 - 2026

- Caught and fixed several critical distributed training bug fixes to open source libraries, e.g. debugged GPU out-of-memory issues due to memory leaks and fixed several DeepSpeed ZeRO-3 integration bugs.

Japanese Grammar Correction LLM | *Keras, Tensorflow*

2021

- Implemented and trained from scratch a **state-of-the-art** grammar correction model from a Grammarly research paper and built a large-scale synthetic data generation pipeline to achieve **+10% accuracy higher than previous SOTA**.

Education

University of Toronto

Toronto, Canada

Bachelors of Science in Computer Science & Linguistics (Honours)

Sep 2015 – Jun 2021

- Studied abroad at University of Tokyo from Sep 2018 - Aug 2019.